Ejercicio 5 (b) - Hoja 1

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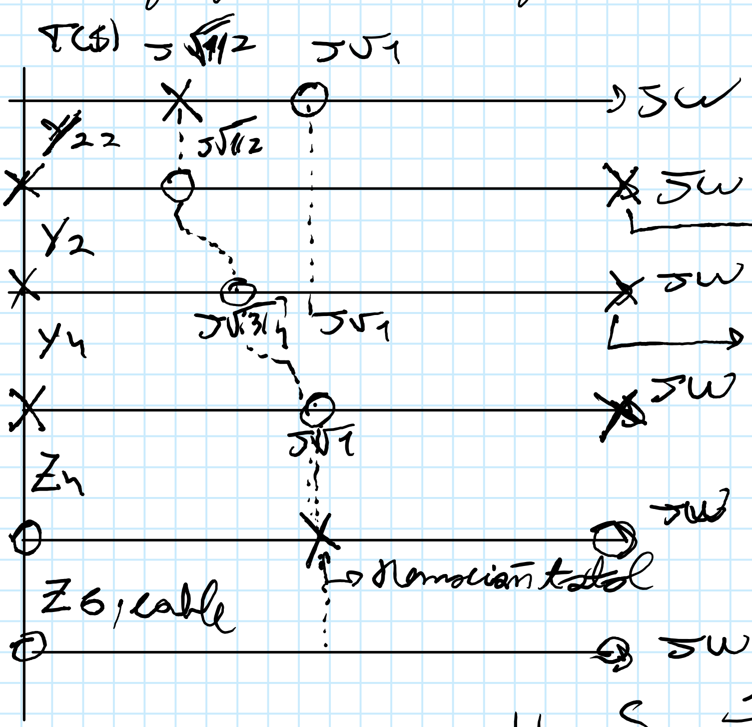
5) Conociendo la transferencia:

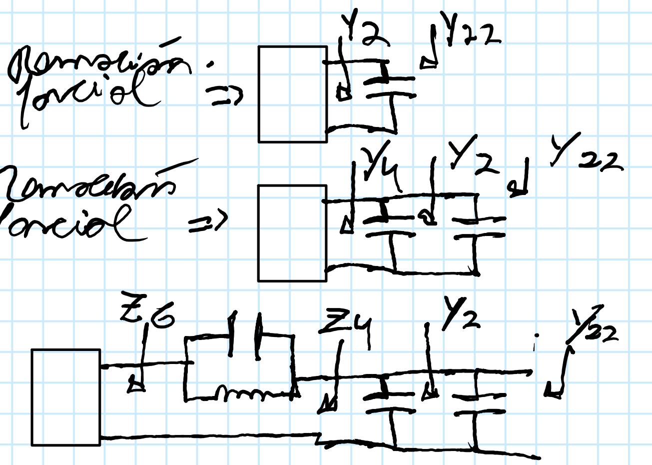
$$\frac{V_2}{V_1} = \frac{s^2+1}{2s^2+1}$$

a) Obtenga un circuito con 3 inductores y 1 capacitor.
b) Obtenga un circuito con 3 capacitores y 1 inductor.

$\frac{V_2}{V_1} = \frac{-Y_{21}}{Y_{22}}$, dando le ceros de Y_{22} producen polos de $T(s)$, recordar $\begin{cases} Y_{22} = \frac{Z_{11}}{Det(Z)} \\ -Y_{21} = \frac{Z_{21}}{Det(Z)} \end{cases}$

1°) Método gráfico de síntesis por remoción





$\frac{1}{s \cdot C} = Z_C$

$s \cdot C = Y_C$

2°) Cálculo de Residuos

- $Y_2(s) = Y_{22}(s) - K'_{\infty,1} \cdot s$, $Y_2(s) = 0 = \left(\frac{2s^2+1}{s} - K'_{\infty,1} \cdot s\right)\Big|_{s=j\sqrt{3/4}}$, $s^2 = -3/4$

$K'_{\infty,1}\, s = \frac{2s^2+1}{s} \Rightarrow K'_{\infty,1} = \frac{2s^2+1}{s^2} \Rightarrow K'_{\infty,1} = \frac{2(-3/4)+1}{(-3/4)} = 2/3$

$K'_{\infty,1} = 2/3$

- $Y_2(s) = \frac{2s^2+1}{s} - (2/3) \cdot s = \frac{(4/3)s^2+1}{s} = Y_2(s)$

- $Y_4(s) = Y_2(s) - K'_{\infty,2} \cdot s$, $Y_4(s) = 0 = \left(\frac{(4/3)s^2+1}{s} - K'_{\infty,2}\, s\right)\Big|_{s=j\sqrt{1}}$, $s^2 = -1$

$K'_{\infty,2} \cdot s = \frac{(4/3) \cdot s^2+1}{s} \Rightarrow K'_{\infty,2} = \frac{(4/3)s^2+1}{s^2} \Rightarrow K'_{\infty,2} = \frac{(4/3)(-1)+1}{-1} = \frac{1}{3}$

$K'_{\infty,2} = 1/3$

$\bullet\, Y_4(s) = Y_2(s) - K_{\infty,2}' \cdot s \,, \quad Y_4(s) = 0 = \left(\frac{(4/3)s^2+1}{s} - K_{\infty,2}' s \right) \Bigg|_{\substack{s=\sqrt{-1} \\ s^2=-1}}$

$K_{\infty,2}' \cdot s = \frac{(4/3) \cdot s^2+1}{s} \Rightarrow K_{\infty,2}' = \frac{(4/3)s^2+1}{s^2} \Rightarrow K_{\infty,2}' = \frac{(4/3)(-1)+1}{-1} = \frac{1}{3}$

$K_{\infty,2}' = 1/3$

Ejercicio 5 (b) - Hoja 2

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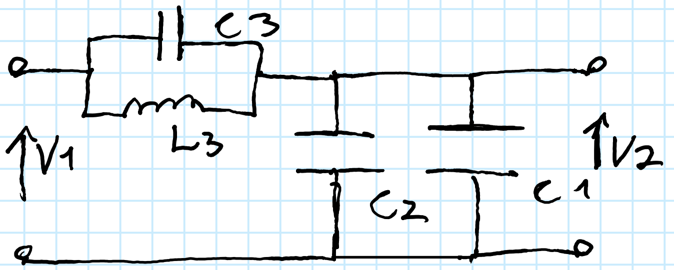
• $Y_4(s) = \frac{(4/3)s^2+1}{s} - (1/3)\cdot s = \frac{s^2+1}{s} = Y_4(s)$, $Z_4(s) = \frac{s}{s^2+1}$

• $Z_6(s) = Z_4(s) - \frac{2K_3\cdot s}{s^2+(\sqrt{1})^2}$, $2K_3 = \lim_{s^2\to -1} \frac{Z_4(s)}{s}\cdot(s^2+(\sqrt{1})^2)$

$2K_3 = \lim_{s^2\to -1} \cdot \frac{\cancel{s}}{\cancel{s^2+1}} \cdot \frac{(s^2+1)}{\cancel{s}} = 1 = 2K_3$

• $Z_6(s) = \frac{s}{s^2+1} - \frac{1\cdot s}{s^2+1} =$ 0 → cable $= Z_6(s)$

3º) Circuito



$C_1 = K'_{\infty,1} = 2/3$

$C_2 = K'_{\infty,2} = 1/3$

$\frac{1}{L_3C_3} = (\sqrt{1})^2 \Rightarrow C_3 = 1$

$L_3 = \frac{1}{2K_3} = 1 = L_3$

4º) Verificación

$V_2 = V_1\cdot \frac{Y_{L_3,C_3}}{Y_{C_1}+Y_{C_2}+Y_{L_3,C_3}}$ ⇒ Divisor de tensión

$Y_{L_3,C_3} = sC_3 + \frac{1}{s\cdot L_3} = \frac{s^2\cdot C_3\cdot L_3 + 1}{s\cdot L_3}\cdot\frac{1/L_3}{1/L_3} = \frac{s^2C_3 + 1/L_3}{s.}$

$$\frac{V_2}{V_1} = \frac{\frac{s^2C_3\cdot L_3+1}{sL_3}}{C_1\cdot s + C_2\cdot s + \frac{s^2C_3L_3+1}{sL_3}} = \frac{\frac{s^2C_3\cdot L_3+1}{\cancel{sL_3}}}{\frac{s^2\cdot L_3(C_1+C_2)+s^2C_3L_3+1}{\cancel{sL_3}}}$$

$$\frac{V_2}{V_1} = \frac{s^2\cdot 1\cdot 1+1}{s^2(1(\frac{2}{3}+\frac{1}{3})+1\cdot 1)+1} = \frac{s^2+1}{2s^2+1} = \frac{V_2}{V_1}$$